Computer Networks

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Evolution of Networking

A computer network or simply a network, is a collection of computers and other hardware components interconnected by communication channels that allow sharing of resources and information.
Computer networks can also be considered a means for transmitting information over long distances.

ARPANET (Advanced Research Projects Agency Network)

In 1969, the U.S. Department of Defence initiated research into joining the computers of defense and research centers into the network. This network became ARPANET, served as starting point for construction of first and most widely known WAN, nowdays known as Internet.

ARPANET joined computers of different types, running various operating system with different add-on modules by implementing communication protocols for all computers participating in network. Such O·S can be considered the first true Network Operating System.

True Network O·S allowed system not only to distribute users but also to organize data storage.

Burst is a sequence of signals counted as one unit.

Bursty traffic refers to uneven pattern of data transmission.

From late 1960s, voice transmission in digital format became increasingly common in telephone network.

ARPANET worked on packets ie packet switching and world's first fully operational packet switching and first network to implement TCP/IP reference model.

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→ It is basically a WAN. It was developed by ARPA. (Advanced Research Project Ajency).

→ It was designed to service even a nuclear attack

→ ARPANET consist of subnet and host computers. Subnet was a datagram subnet and each subnet consist of minicomputers called IMPs (Interface Message Processors). Host could send messages of upto 8063 bits to its IMP which would break them into packets are forward them independently toward destination.

Internet
is a global network of billion of computers and other electronic devices.
Web - world wide web (www) is a collection of different websites you can access through internet. When you visit a website, your computer sends a request over these wires to server. A sever is where websites are stored. once a request arrives, the server retrieves the website and sends correct data back to computer. WWW is biggest service of Internet.

Types of Internet Service.
The type of Internet service you choose will depend on Internet service provider (ISPs) serve by your area.

(1) Dial-up: slowest type of Internet connection. It uses your phone line ie landline phone at home.

(11) DSL: DSL service uses broadband connection, faster than Dialup, connect Internet via phoneline but doesnot require landline at home.

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(iii) Cable: connects via cable TV, uses broadband connection, faster than both dial-up and DSL service.

(iv) Satellite: it connects to internet through satellites orbiting the Earth, can be affected by weather pattern, slower than DSL or cable.

(v) 3G or 4G: connects wirelessly through ISP network. they limit amount of data for every month, not always fast as DSL or cable.

Internet basics

(i) Internet and www are not same thing.

(ii) utilizes TCP/IP protocol and is accessed using modem, broadband, 3G, 4G, or network connected through an ISP.

(iii) Internet is explored (called surfing) using browser.

(iv) Find information on Internet is achieved using search engine

Interspace

is basically a client/server software program, allows multiple users to communicate online with real time audio, video or text chat in dynamic 3D environment.

Interspace provides most advanced form of communication available on Internet today.

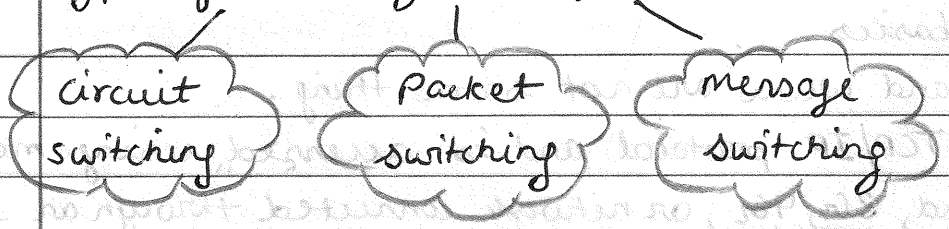
In short, Interspace is a vision of what Internet will become. It is an applications environment for interconnecting spaces to manipulate information.

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Introduction

Aim of data communication and networking is to facilitate exchange of data such as audio, text or video b/w various points in world. For delivery of data or information with ease of accuracy various types of Switching techniques are employed in Data communication & Networking.

Types of Switching Techniques.



I) Circuit switching.

Generally used in public networks.

- circuit switching is a technique that directly connects the sender and receiver in an unbroken path. ie. dedicated communication path.
- When A needs to communicate with B, system A needs to request a connection to system B that must be accepted by all switches as well as by B itself. This is called setup phase. Once a connection is established, a dedicate path exists b/w both ends until the connection is terminated.
- Routing decision must be made when circuit is first established, but there are no decision made after that time.
- a complete end-to end path must exist before communication can take place.

advantage - Communication channel (once established) is dedicated.

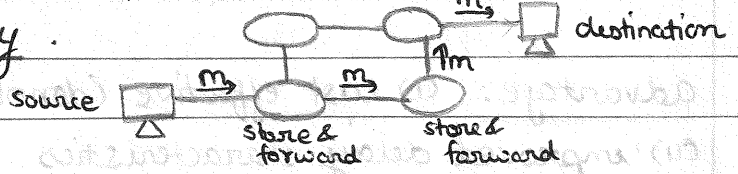
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Disadvantage.

→ possibly long wait to establish a connection
→ more expensive than any other switching technique because of dedicated path.
→ inefficient use of communication channel, because channel is not used when connected system are not using it.

Message switching.

- not necessary to establish a dedicated path b/w sender and reciever. Each message is routed independently through the network.
- when a station sends a message, destination address is apperded to message.
- message is transmitted through network from node to node.
- Each nodes recieves a message, stores it in its entirely on disk, and then transmits message to next station. ie why called store and forward network.
- It is typically a general purpose computer. Device needs sufficient storage capacity to store.. forward time, plus time required to find next node in transmission path leads to time delay.



advantage:

i) channel efficient is greater, bcz more devices are sharing channel.
ii) traffic congestion can be reduced due to store in route
iii) message priorities due to store and forward network
iv) message broadcasting can be achieved due to address apperded in message.

disadvantages: (i) not compatible with interactive applications.
(ii) store and forward are expensive because they must have large disk to hold long messages.

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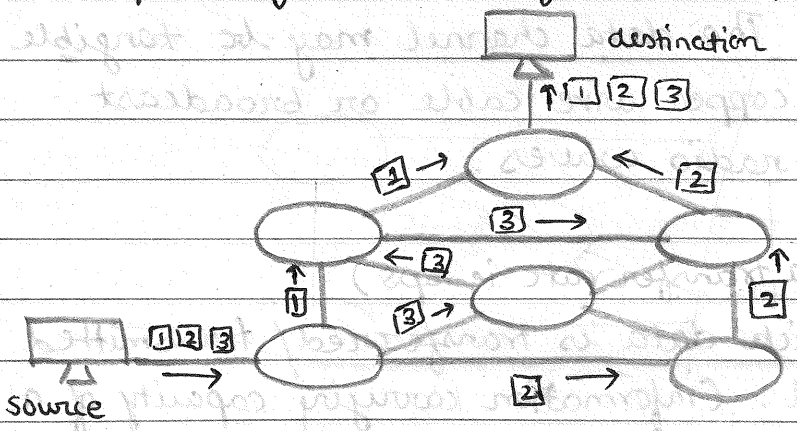
Packet switching → Datagram P.S
→ Virtual Circuit P.S

- messages are divided into packets of fixed or variable size. It combines advantage of message and circuit switching and to minimize disadvantage of both.
- size of packet is decided by network and governing protocal.
- resources are allocated on demand. Resource allocation is done on first come - first serve basis.
- Each switching node has small amount of buffer space to hold packets temporarily.
- If outgoing line is busy, packet stays in queue until line become available.
- Each packet has source and destination addresses.
- Since packets have a strictly defined length, they can be stored in mainmemory instead of disk.
- In packet switching, analog signal from your phone is converted into digital data stream.
- packets donot travel together, sequentially, or same route.

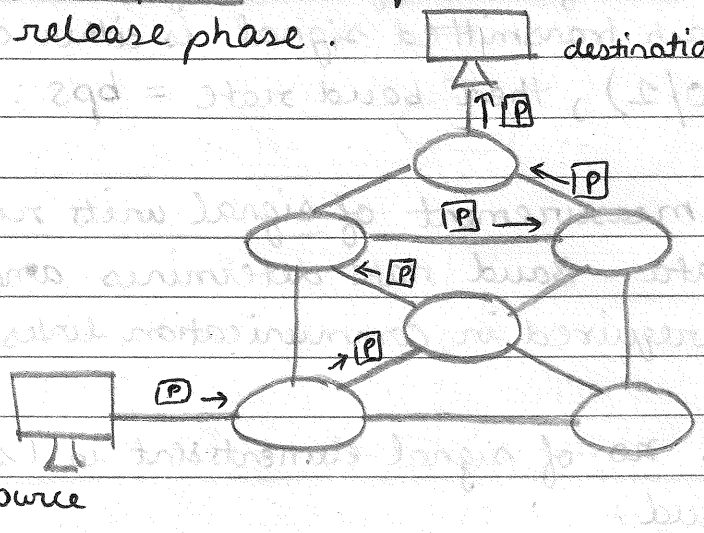
advantage: (i) cost effective (donot need massive storage)
(ii) improved delay characteristics (no long messages)
(iii) packets can be rerouted (if busy or disable links)
(iv) many network can share channel at same time

disadvantages: (i) protocols are typically more complex
(ii) add some initial cost in implementation.
(iii) if packet is lost, sender needs to retransmit data.
(iv) cannot deliver same quality as dedicate circuit

(i) Datagram Packet switching.
↳ normally implemented in network layer.
↳ each packet is routed independently through network.
↳ each packet has header that contains destination address.
↳ when switch recieves the packet, destination address in header is examined, routing table is consulted to find corresponding port through which packet should be forwarded.



(ii) Virtual circuit packet switching.
↳ normally done at data link layer
↳ fixed path b/w sender & receiver (also called connection oriented)
↳ three phase: setup phase, data transfer phase, connection release phase.



↳ node doesnot make routing decision for each packet.
↳ it implies acknowlegment, flow control & error control. so virtual circuits are reliable.

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Data communication terminologies.

(I) Channel.

Physical medium like cables over which information is exchanged is called channel. Transmission channel may be analog or digital.

Path over which data is sent or received is called data channel. The data channel may be tangible medium like copper wire cable or broadcast medium like radio waves.

(II) Baud.

Speed at which data is transferred/ transmitted over a channel. (Information carrying capacity of a channel). deals with signals.

At low speeds, one baud is equal to 1bps

$\Rightarrow$ 1200 baud channel will transmit data at a rate of 1200 bits per second.

Baud indicates no. of line signal changes per second Therefore, if each transmitted signal is either a zero or one (0/1), then baud rate = bps.

Baud rate is measurement of signal units required to transmit data. Baud rate determines amount of bandwidth required in communication links

Signal rate: is no. of signal elements sent in 1 sec. The unit is baud.

Signal rate is sometimes called pulse rate, modulation rate or baud rate.

Baud can never be greater than data transfer rate

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Data rate

no. of data elements (bits) sent in 1 second. The unit is bps. (bits per second).

It is used to compare speeds and transmission

1 Bps = 8 bits per second

↳ byte per second.

(2^{10})

1 Kbps = 1 kilobits per second = 1024 bits per second.

1 Mbps = 1 Megabits per second = 1024 Kbps = 2^{20} bps

1 Gbps = 1 Gigabits per second = 1024 Mbps = 2^{30} bps.

1 Tbps = 1 Terabits per second = 1024 Gbps = 2^{40} bps.

Note: 'B' stands for Byte per second
'b' stands for bits per second.

Bandwidth.

difference b/w lowest and highest frequencies of a transmission channel. (ie. allocated band of frequencies to a channel).

Bandwidth is directly propotional to amount of data transmitted or recieved per unit time.

High bandwidth channels are called broadband channel and low bandwidth channels are called narrowband channel.

In digital, bandwidth is data speed in bps. ie (bandwidth in bits per second)

In analog, bandwidth is termed in frequency ie. cycle per second ie hertz. (bandwidth in hertz).

↳ KHz, MHz, GHz, THz → a thousand GHz. (10^9 to 10^{12})

thousand cycle per second. (10^{-1})

a thousand KHz (10^{+3})

a thousand MHz (10^6)

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Transmission Mode.

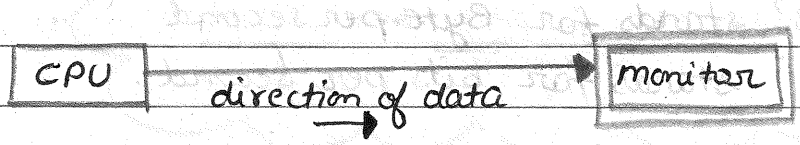
↳ refers to mechanism of transferring of data b/w two devices connected over a network. (communication mode). These modes directs the direction of flow of information.

(i) Simplex mode.

↳ data can be sent only in one direction ie unidirectional.

↳ We just need to send a command / signal and do not ~~except~~ expect any response back.

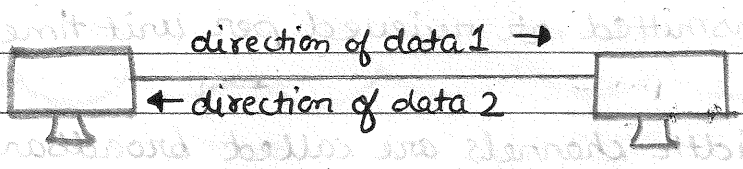
↳ example: loudspeakers, television broadcasting, TV remote, keyboard and monitor etc.



(ii) Half duplex mode.

↳ data can be transmitted in both direction on a signal, but not at same time.

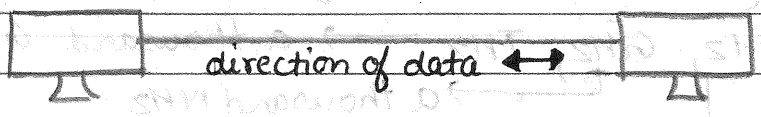
example: walkie-talkie in which messages are sent one at a time but in both directions.



(iii) Full duplex mode.

↳ data can be transmitted in both directions at same time (bidirectional)

example: telephone conversation.



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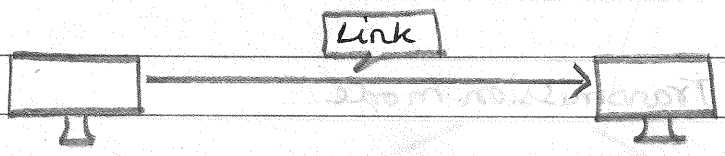
Line Configuration

↳ a network is nothing but a connection made through connection links b/w two or more devices.

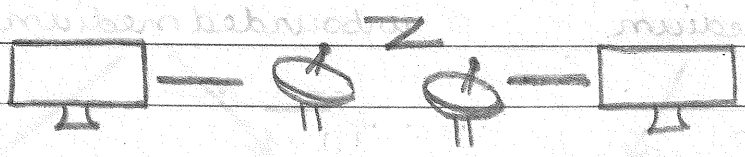
(i) **Point to Point Connection. (PPP)**

↳ protocol which is used as a communication link between two devices. Simple to establish.

↳ most common example: (PPP) is computer connected by telephone lines.



can be connected by wires or microwave / satellite link

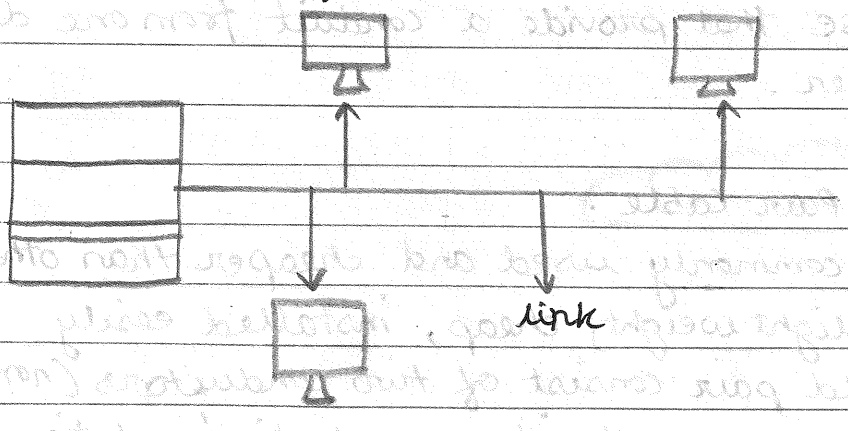


(ii) **Multipoint Connection. (Multidrop configuration)**

↳ two or more devices share a single link.

↳ if links are used simultaneously b/w many devices, then it is shared line configuration.

↳ if user takes turn while using link, then it is time shared line configuration.



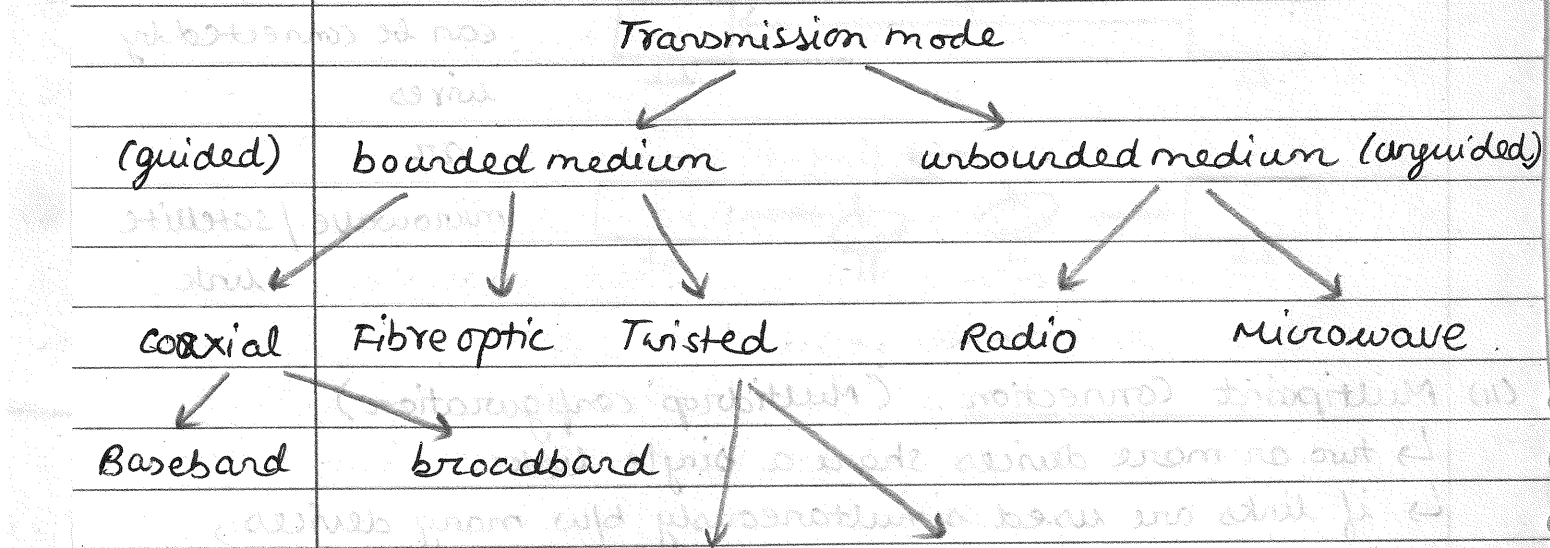
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Transmission medium in computers.

→ Signals are transmitted in form of electromagnetic energy from one device to another. They can travel through vacuum, air or other transmission medium.

→ Electromagnetic energy consist of power, voice, visible light, radio waves, ultraviolet light, gamma rays etc.

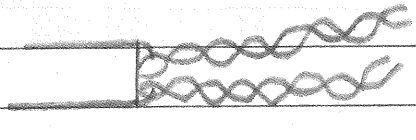
→ The first layer (Physical layer) is dedicated to transmission media.



(1) Bounded (Guided) Transmission Media.
are those that provide a conduit from one device to another.

Twisted Pair Cable.

- most commonly used and cheaper than others, it is light weight, cheap, installed easily
- twisted pair consist of two conductors (normally copper), each with its own plastic insulation, twisted together.



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→ one of wire is used to carry signal to reciever and other is used only as ground reference. The receiver uses difference b/w the two. (to decrease noise (interference) and crosstalk).

Unshielded Twisted Pair Cable.

↳ consist of two conductors usually copper, each with its own colour plastic insulator. (due to identification).

↳ UTP cables consist of 2 or 4 pairs of twisted cable.

↳ Cable with 2 pair use RJ-11 connector
4 pair use RJ-45 connector.

advantage: (i) Installation is easy
(ii) Flexible and easy
(iii) cheap and high speed capacity
(iv) 100 m limit

disadvantage: (i) bandwidth is low when compared with coaxial cable.
(ii) less protection from interference.

Shielded Twisted Pair Cable.

↳ cable has a metal foil or braided-mesh covering which encases each pair of insulated conductors. (electromagnetic noise penetration is prevented by metal casing).

↳ It is faster than unshielded and coaxial ie, why more expensive.

advantage: (i) easy to install
(ii) can be used for analog and digital transmissi
(iii) increases signal rate., eliminates crosstalk.

disadvantage: (i) difficult to manufacture
(ii) heavy.

applications of STP.

(i) In telephone lines (DSL lines)

(ii) Local Area Network, such as 10 Base-T, 100 Base-T, also use twisted pair cable.

Coaxial cable:

↳ contains two conductor that are parallel to each other. (copper is used at center).

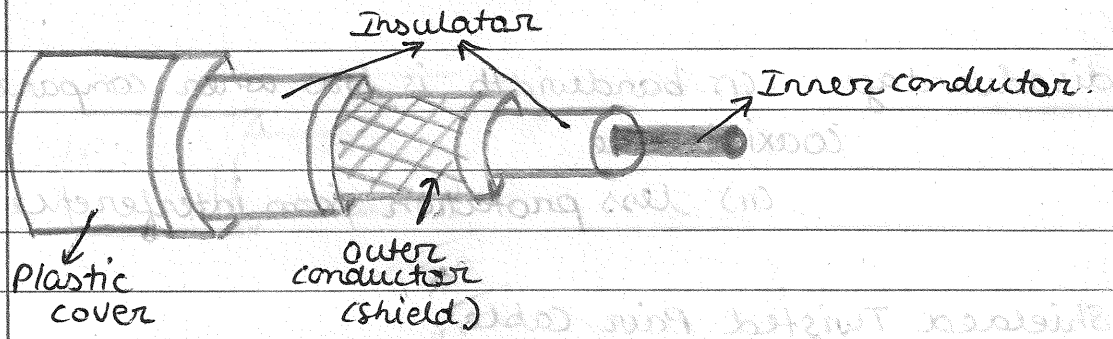
Most common coaxial standard:

50-ohm RG-7 or RG-11: thick Ethernet

50-ohm RG-58: thin Ethernet

75-ohm RG-59: cable television

93-ohm RG-62: ARCNET



Coaxial cables are categorized by their Radio Government (RG) ratings.

Imp →

RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet.

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Coaxial cable connectors.

- BNC Connector (cable to device) ex TV.
- BNC T connector (Ethernet network)
- BNC terminator. (end of cable to prevent reflection of signal).

BNC stands for Bayonet Neill-Concelman.

Coaxial cables → Baseband
Coaxial cables → Broadband

	Baseband	Broadband
(i)	digital transmission	(i) analog transmission
(ii)	transmits single signal at a time with very high speed	(ii) transmits several simultaneous signal using different frequencies.
(iii)	needs amplification after every 1000 feet.	(iii) covers large area.

advantage:
(i) bandwidth is high
(ii) used in long distances, high rate of 10 Mbps
(iii) much higher noise immunity.
(iv) data transmission without distortion.

disadvantages:
(i) single cable failure can fail entire network.
(ii) difficult to install & expensive.
(iii) if shield is imperfect, it can lead to grounded loop.

10 Base-2 is called Thin Ethernet.

10 Base-5 Thick Ethernet, Thicknet, Thickwire

10 Base-T thinnest among all — RJ-45 connector is connected with.

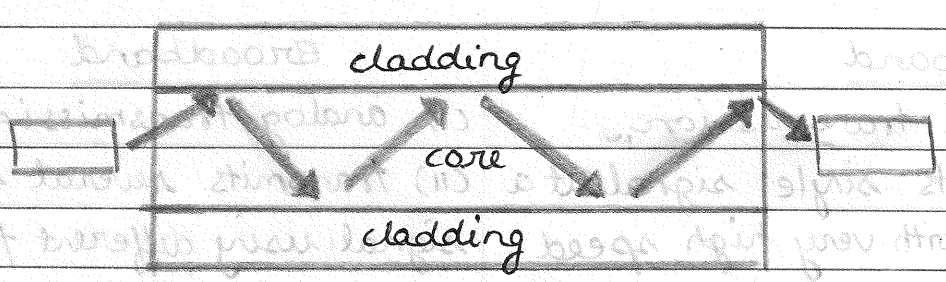
Fibre Optic cable.

L is made of glass or plastic and transmits signal in form of light.

L it uses reflection to guide light through a channel.

L A glass or plastic core is surrounded by a cladding of less dense glass or plastic. The difference in density of two material must be such that a beam of light moving through core is reflected off the cladding instead of being refracted into it.

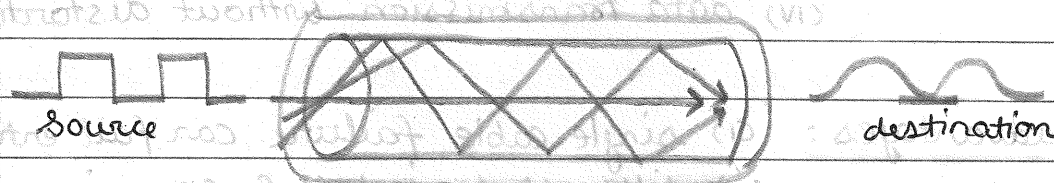
total internal reflection



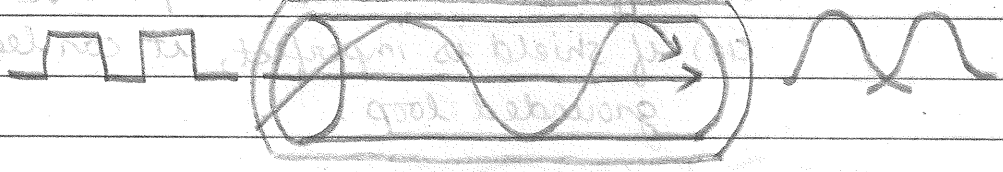
Propogation mode of fibre optic cable.

- Multimode
  - Step index
  - graded index
- single mode

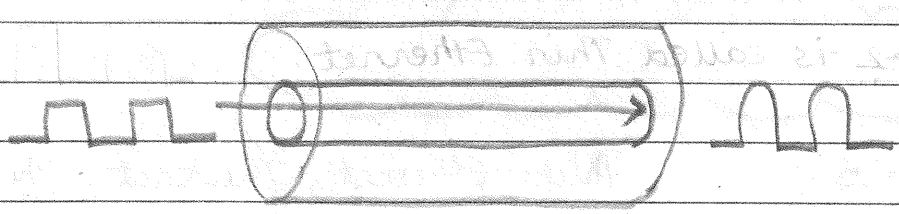
Multimode step index



multimode graded index



single mode



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In multimode step index → density remains constant from centre to edges of core.

graded index → distortion gets decrease through cable. (index refers to index of refraction). density varies. Density is highest at center of core and decreases gradually at edges.

Single mode → uses step index fibre. It is manufactured with much small diameter than multimedia with substantially lower density.

Fibre optic connectors.

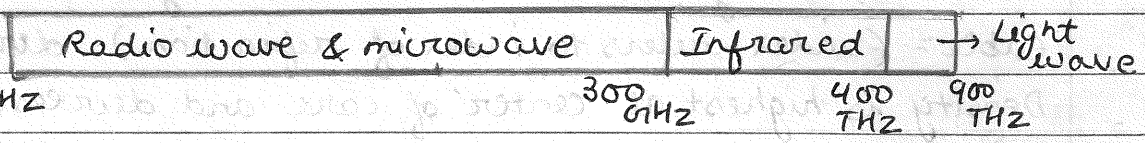
- SC (subscribe channel) connector
  - ↓ cable TV
  - (push/pull locking system)
- ST (straight-Tip) connector
  - ↓ cable to networking device
- MT-RJ connector
  - ↓ same as RJ-45

advantages: (i) higher bandwidth, less signal attenuation
(ii) light weight, immunity to electromagnetic interference
(iii) resistance to corrosive material.

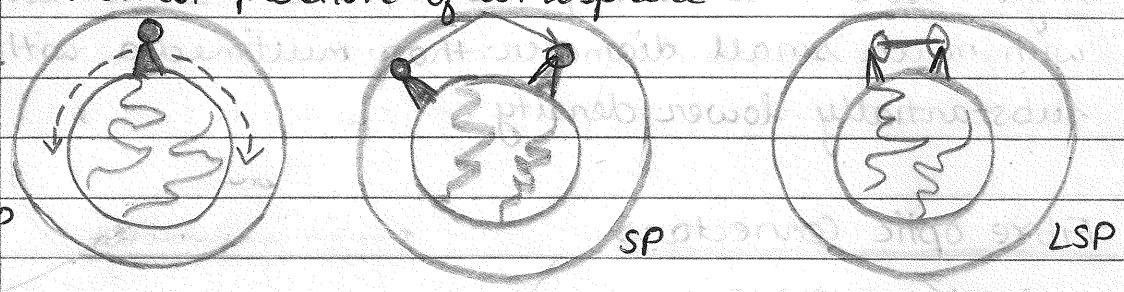
disadvantages: (i) installation & maintenance
(ii) unidirectional
(iii) high cost.

application: Local area Network such as 100 Base-FX and 1000 Base-X network

(II) Unbounded (Unguided) transmission
↳ without using physical conductor. (wireless)



GP- Ground propogation: (below 2 MHz)
↳ lowest portion of atmosphere



SP- Sky propogation (2-30 MHz)
↳ higher frequency radio waves radiate upward into ionosphere where they are reflected back).

LSP - Line of sight Propogation (above 30 MHz)
↳ very high frequency signals are transmitted in straight line directly from antenna to antenna.

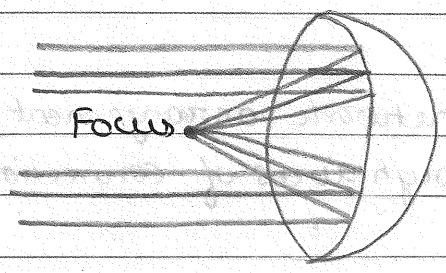
We can divide wireless transmission into three broad groups:

(I) Radio wave.
↳ are omnidirectional. ie. receiving and sending antennas do not have to be aligned.
↳ omnidirectional property has disadvantage too as it is susceptible to interference by another antenna that may send signal using same frequency/band.
↳ radio waves with low & medium frequency can penetrate walls.

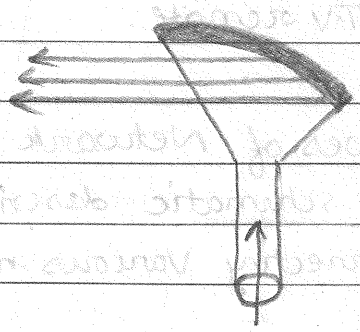
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application of radio waves
↳ AM and FM radio, television, maritime radio, cordless phones, paging are example of multicasting.

(ii) Micro waves.
↳ are unidirectional ie sending and receiving antennas need to be aligned.
↳ they are line-of-sight.
↳ very high-frequency microwave cannot penetrate walls
↳ use of certain portions of band require permission from authorities.



(i) Dish antenna

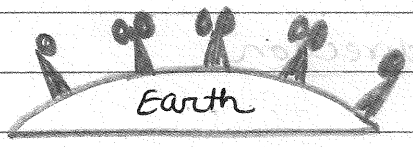


(ii) horn antenna.

application of micro wave.
used in cellular phones, satellite network & wireless LANs.

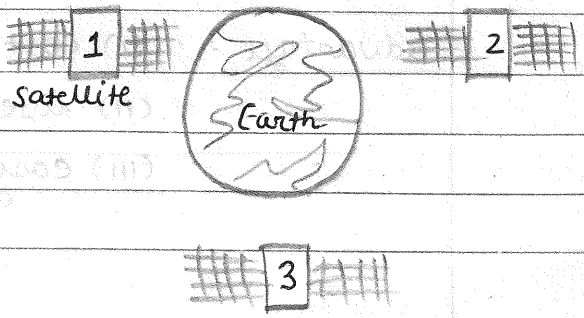
There are 2 types of Microwave Transmission

(i) Terrestrial Microwave
(a) for increasing distance, repeaters are installed with every antenna.



Signal received by antenna can be converted into transmittable form and relayed to next antenna.

(ii) Satellite microwave.
placed in outer space, positioned above 36000km above equator that matches exactly rotation speed of earth.



(iii) Infrared waves.

- short range communication
- infrared rays having high frequency cannot penetrate walls.

application

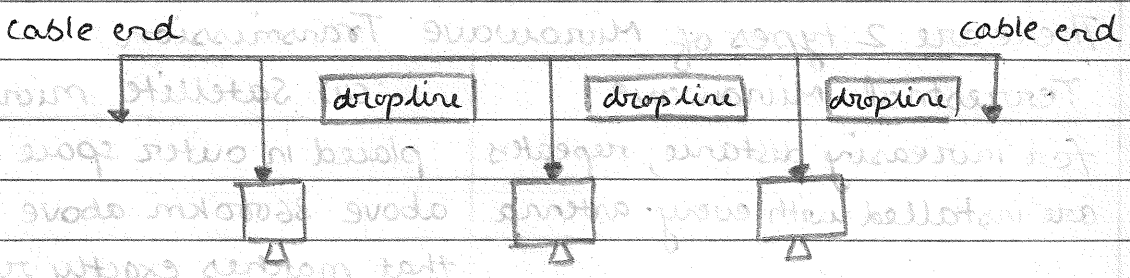
- used to transmit digital data with very high data rate.
- use signals for keyboard, mouse, PCs, printers
- line of sight communication (short range) like TV remote.

Types of Network Topology.

is schematic description of network arrangement, connecting various nodes through lines of connection.

Bus Topology.

every computer and network device is connected to single cable. When, it has exactly two endpoints, then it is called Linear Bus Topology.



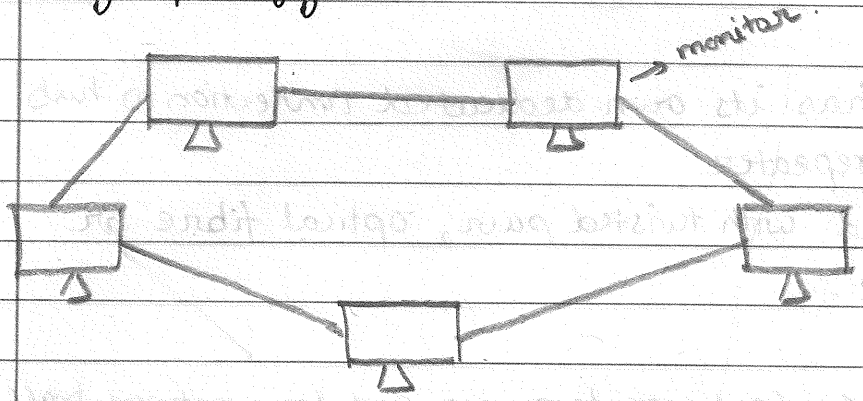
→ transmits data only in one direction

advantage: (i) cost effective, cable required is least
(ii) used in small network.
(iii) easy to expand.

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disadvantage: (i) cable fails whole network fails
(ii) cable has limited length.
(iii) slower than ring topology.
(iv) if nodes are more, performance decreases.

Ring topology.



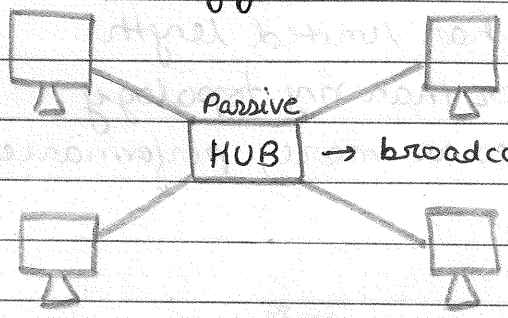
- a no. of repeaters are used for ring topology with large no nodes
- transmission is unidirectional but can be made bidirectional by having 2 connections (dual ring topology). Data flows in opp direction in them. If one ring fails, second ring acts as a backbone.
- data transmitted has to pass through each node in network.

advantages: (i) transmitting network is not affected by high traffic or adding more no. of nodes, as only nodes having token can transmit data.
(ii) cheap to install and expand.

disadvantage: (i) troubleshooting is difficult
(ii) adding or deleting node disturbs network
(iii) failure of one node disturbs whole network.

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Star topology



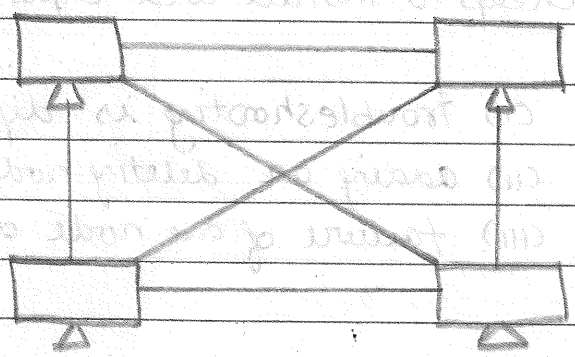
→ broadcast device / browsing center / nonintelligent device

- every node has its own dedicated connection to hub
- hubs acts as repeater
- can be used with twisted pair, optical fibre or coaxial cable.

advantage: (i) fast performance and low network traffic
(ii) hub can be upgraded easily
(iii) easy to troubleshoot, setup, modify
(iv) if one node fails, rest system is not affected.

disadvantage: (i) cost of installation is high
(ii) expensive
(iii) If hub fails then whole network fails.

Mesh topology



- point to point connection to other nodes.
- all nodes are connected to each other.
- fully connected, robust, not flexible.

There are two techniques in Mesh topology

Routing Flooding

(i) **Routing.**
- nodes have a routing logic to direct data to reach the destination using shortest distance.
- routing logic has information about broken links

(ii) **Flooding.**
- no routing logic is required, same data is transmitted to all nodes
- network is robust, and is likely to loose data, leads to unwanted load over network.

Types of Mesh topology.

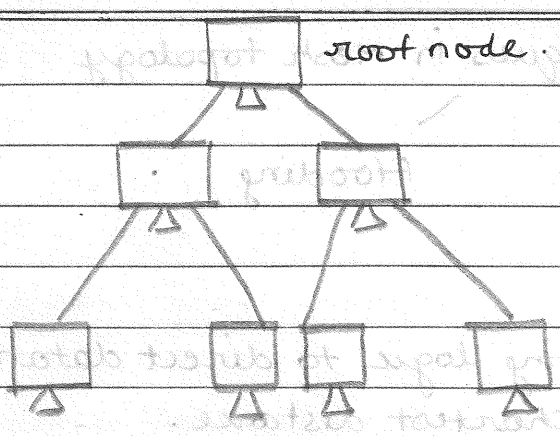
(i) Partial: some devices are only connected to two or three devices

(ii) Full: Each and every device are connected to each another

advantages: (i) each connection carry its own data load.
(ii) robust and fault can be diagnosed easily
(iii) secure and privacy

disadvantage: (i) installation & configuration is difficult.
(ii) cable cost is more as bulk wiring is required.

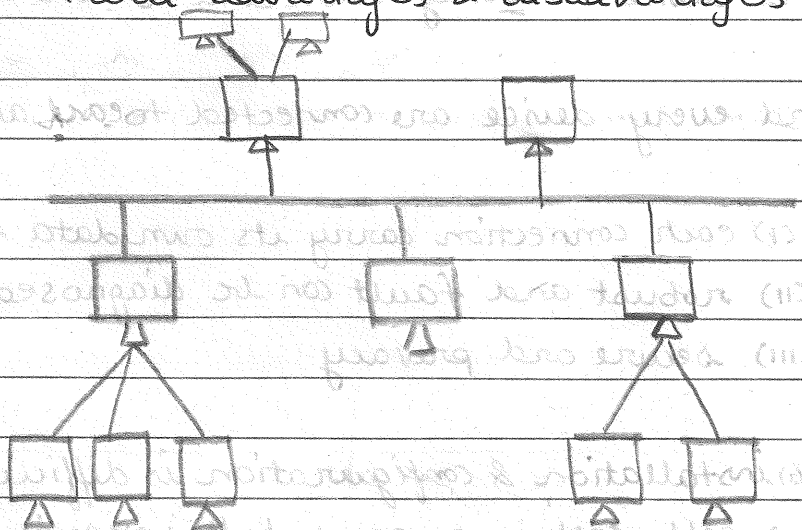
Tree topology (also called hierarchical topology)
- it has root node and all other nodes are connected to it.
- ideal if workstation are located in group.
- used in WAN.



advantage: (i) extension of bus and star topology
(ii) expansion of node is easy
(iii) error detection is easily done.

disadvantage: (i) heavily cabled, costly
(ii) central hub fails, network fails.

Hybrid technology.
↳ mixture of two or more topologies.
↳ inherit advantages & disadvantages of topologies.



advantage: (i) error detecting & troubleshoot is easy
(ii) effective, flexible
(iii) scalable as size can be increased easily

disadvantage: (i) complex in design
(ii) costly.